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Tutorial Course for Intro to Sensor-Integrated Systems

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Measurement Error Basics

1. Quantitative Definitions of Errors

- **Absolute Error:** $\Delta x = x - x_0$ (Measured Value - True Value)
- **Relative Error:** $\gamma = \frac{\Delta x}{x_0} \times 100\%$ (Reflects measurement accuracy more objectively)
- **Referenced / Fiducial Error:** $\gamma_n = \frac{\Delta x}{x_{F,S}} \times 100\%$ (Relative to the Full Scale $x_{F,S}$; used to define instrument accuracy classes)

2. The Three Types of Errors & Countermeasures

- **Systematic Error:**
 - *Characteristics:* Remains constant or changes according to a specific pattern.
 - *Countermeasure:* **Cannot** be eliminated by repeated measurements! Requires finding the root cause or introducing a "correction value" for compensation.
- **Random Error:**
 - *Characteristics:* Unpredictable in magnitude and sign, but follows statistical laws (e.g., Normal Distribution) over multiple trials.
 - *Countermeasure:* Increase the number of measurements and calculate the average (mean).
- **Gross Error / Outlier:**
 - *Characteristics:* Extreme data points that significantly distort the results (usually caused by human mistakes or sudden instrument faults).
 - *Countermeasure:* Identify and strictly eliminate using statistical criteria like the 3σ rule or Grubbs' test.

Core Data Processing Methods

1. Statistical Description of Data

- **Arithmetic Mean:** $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ (The best estimate of the true value)
- **Residual Error:** $v_i = x_i - \bar{x}$
- **Standard Deviation (Single Measurement):** $\sigma \approx S = \sqrt{\frac{\sum v_i^2}{n-1}}$ (Bessel's formula; evaluates the uncertainty or dispersion of data)
- **Weighted Average:** Used when datasets have different precisions. The weight is inversely proportional to the variance: $p_i = \frac{1}{\sigma_i^2}$.

2. How do Errors "Propagate"? (Indirect Measurement)

- Given a functional relationship $y = f(x_1, x_2)$ with independent variables.

- **Root-Sum-Square (RSS) Formula:** $\sigma_y = \sqrt{\left(\frac{\partial f}{\partial x_1} \sigma_{x_1}\right)^2 + \left(\frac{\partial f}{\partial x_2} \sigma_{x_2}\right)^2}$

3. Method of Least Squares (Linear Regression)

- *Core Idea:* Find the best-fit straight line $y = a + bx$.
- *Objective:* Minimize the sum of the squared deviations of all measured points from the line: $Q = \sum_{i=1}^n (y_i - (a + bx_i))^2 = \min$.
- *Application:* The ultimate tool for determining a sensor's static calibration, including its **Sensitivity** (slope b) and **Zero Offset** (intercept a).

Exercise

Measurements of the speed of light were conducted, yielding four sets of results along with their standard deviations as follows:

- **Set 1:** $C_1 = 2.98000 \times 10^8$ m/s, $\sigma_1 = 0.01000 \times 10^8$ m/s
- **Set 2:** $C_2 = 2.98500 \times 10^8$ m/s, $\sigma_2 = 0.01000 \times 10^8$ m/s
- **Set 3:** $C_3 = 2.99990 \times 10^8$ m/s, $\sigma_3 = 0.00200 \times 10^8$ m/s
- **Set 4:** $C_4 = 2.99930 \times 10^8$ m/s, $\sigma_4 = 0.00100 \times 10^8$ m/s

Task: Calculate the **weighted arithmetic mean** (加权算术平均值) of the speed of light and its **standard deviation** (标准差).

Solution

$$P_1:P_2:P_3:P_4 = \frac{1}{\sigma_{\bar{x}_1}^2}:\frac{1}{\sigma_{\bar{x}_2}^2}:\frac{1}{\sigma_{\bar{x}_3}^2}:\frac{1}{\sigma_{\bar{x}_4}^2} = 1:1:25:100$$

$$\bar{x}_{\bar{p}} = \frac{(2.98000 \times 1 + 2.98500 \times 1 + 2.99990 \times 25 + 2.99930 \times 100) \times 10^8}{1 + 1 + 25 + 100} = 2.99915 \times 10^8 \text{ m/s}$$

$$\sigma_{\bar{x}_{\bar{p}}} = \sqrt{\frac{1 \times (2.98000 - 2.99915)^2 + 1 \times (2.98500 - 2.99915)^2 + 25 \times (2.99990 - 2.99915)^2 + 100 \times (2.99930 - 2.99915)^2}{(4-1)(1+1+25+100)}}$$
$$= 0.00127 \times 10^8 \text{ m/s}$$

Exercise

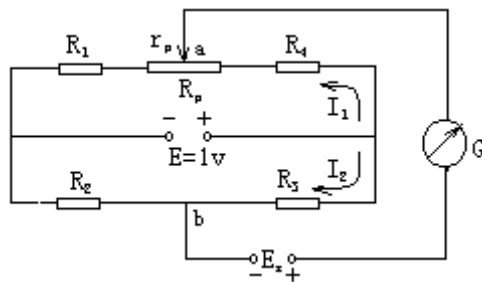
A potentiometer circuit is used to measure an unknown electromotive force (EMF) signal, E_x .

The known parameters of the balanced circuit are:

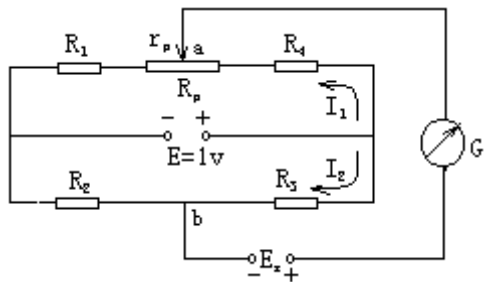
- Branch currents: $I_1 = 4 \text{ mA}$, $I_2 = 2 \text{ mA}$
- Resistances: $R_1 = 5\Omega$, $R_2 = 10\Omega$, $R_p = 10\Omega$, and the active sliding portion $r_p = 5\Omega$.
- The known constant systematic errors (absolute errors) of the resistors are:
 $\Delta R_1 = +0.01\Omega$, $\Delta R_2 = +0.01\Omega$, $\Delta r_p = +0.005\Omega$.

Assuming the errors of the galvanometer and the stable branch currents (I_1 , I_2) are negligible.

Calculate the true magnitude of E_x after eliminating the systematic errors.



Solution



$$I_1(R_1 + r_p) - I_2 R_2 = E_x$$

$$E_x = I_1(R_1 + r_p) - I_2 R_2 = 4 \times (5 + 5) - 2 \times 10 = 20\text{mv}$$

$$\Delta E_x = \frac{\partial E_x}{\partial R_1} \Delta R_1 + \frac{\partial E_x}{\partial r_p} \Delta r_p + \frac{\partial E_x}{\partial I_1} \Delta I_1 + \frac{\partial E_x}{\partial R_2} \Delta R_2 + \frac{\partial E_x}{\partial I_2} \Delta I_2$$

$$= I_1 \Delta R_1 + I_1 \Delta r_p + R_1 \Delta I_1 + r_p \Delta I_1 - I_2 \Delta R_2 - R_2 \Delta I_2$$

$$= 4 \times 0.01 + 4 \times 0.005 - 2 \times 0.01 = 0.04\text{mv}$$

$$E'_x = 20 - 0.04 = 19.96\text{mv}$$

Exercise

The current and voltage of a specific circuit are measured as $I = 22.5 \text{ mA}$ and $U = 12.6 \text{ V}$.

Their corresponding standard deviations (uncertainties) are $\sigma_I = 0.5 \text{ mA}$ and $\sigma_U = 0.1 \text{ V}$, respectively.

Task: Calculate the nominal consumed power P and its standard deviation σ_P .

Solution

The current and voltage of a specific circuit are measured as $I = 22.5 \text{ mA}$ and $U = 12.6 \text{ V}$.

Their corresponding standard deviations (uncertainties) are $\sigma_I = 0.5 \text{ mA}$ and $\sigma_U = 0.1 \text{ V}$, respectively.

Task: Calculate the nominal consumed power P and its standard deviation σ_P .

$$P_0 = UI = 22.5 \times 12.6 = 283.5 \text{ mW}$$

$$\sigma = \sqrt{U^2 \sigma_I^2 + I^2 \sigma_U^2} = \sqrt{12.6^2 \times 0.5^2 + 22.5^2 \times 0.1^2} = 6.69 \text{ mW}$$

Exercise

When using an X-ray machine to inspect internal defects of magnesium alloy castings, the penetrating voltage (y) should vary with the thickness of the inspected part (x) to achieve optimal sensitivity. An experiment yielded the following dataset.

Task: Determine the best-fit empirical formula for the penetrating voltage (y) as a function of thickness (x) using the **Method of Least Squares**.

X/m	12	13	14	15	16	18	20	22	24	26
m										
Y/kv	52. 0	55. 0	58. 0	61. 0	65. 0	70. 0	75. 0	80. 0	85. 0	91. 0

Solution

Assuming the scatter plot indicates a linear relationship, we define the empirical formula as:

$$y = a + bx$$

According to the Method of Least Squares, the optimal coefficients a and b must minimize the

sum of squared residual errors. This yields the following system of linear normal equations:

$$1. \quad n \cdot a + b \sum x_i = \sum y_i$$

$$2. \quad a \sum x_i + b \sum x_i^2 = \sum x_i y_i$$

Given the number of data points $n = 10$, we calculate the sum components from the table:

- $\sum x_i = 12 + 13 + \cdots + 26 = \mathbf{180}$
- $\sum y_i = 52.0 + 55.0 + \cdots + 91.0 = \mathbf{692}$
- $\sum x_i^2 = 12^2 + 13^2 + \cdots + 26^2 = \mathbf{3450}$
- $\sum x_i y_i = (12 \times 52) + (13 \times 55) + \cdots + (26 \times 91) = \mathbf{13032}$

Substitute the summations into the normal equations:

$$1. \quad 10a + 180b = 692$$

$$\mathbf{b \approx 2.74}$$

The best-fit empirical formula for the penetrating voltage is:

$$2. \quad 180a + 3450b = 13032$$

$$\mathbf{a \approx 19.88}$$

$$\mathbf{y = 19.88 + 2.74x}$$

Sensor Static Characteristics

1. Core Definitions (Key Indicators)

- **Sensitivity (S):** The ratio of the incremental output (Δy) to the incremental input (Δx), i.e., $S = \Delta y / \Delta x$.
- **Linearity (Non-linearity):** The maximum deviation of the actual calibration curve from a specified fitted straight line.
- **Hysteresis:** The non-coincidence between the loading (increasing input) and unloading (decreasing input) output curves of a sensor.
- **Repeatability:** The closeness of agreement among successive measurements of the same input, carried out in the same direction under identical conditions.
- **Drift:** The variation in the sensor's output over time or due to environmental changes (e.g., temperature) while the input remains constant.

2. Quantifying Errors (Calculation Formulas)

- **Linearity Error:** $\gamma_L = \pm \frac{\Delta L_{max}}{y_{F.S}} \times 100\%$
 - ΔL_{max} : Maximum non-linear deviation; $y_{F.S}$: Full-scale (F.S.) output.
- **Hysteresis Error:** $\gamma_H = \pm \frac{\Delta H_{max}}{y_{F.S}} \times 100\%$
 - ΔH_{max} : Maximum output difference between the loading and unloading curves.
- **Repeatability Error:** $\gamma_R = \pm \frac{k \cdot \sigma}{y_{F.S}} \times 100\%$
 - σ : Standard deviation of the measurements; k : Confidence coefficient (usually 2 or 3, depending on the required confidence level).

Exercise

A pressure sensor was tested over three calibration cycles. The input pressure (x , in MPa) and the corresponding output voltage (y , in mV) for both the **loading** (forward stroke) and **unloading** (reverse stroke) processes are recorded in the table below.

Task: Calculate the **Non-linearity Error**, **Hysteresis Error**, and **Repeatability Error** of this pressure sensor.

Raw Calibration Data:

Pressure (MPa)	Cycle 1 Loading	Cycle 1 Unloading	Cycle 2 Loading	Cycle 2 Unloading	Cycle 3 Loading	Cycle 3 Unloading
0.00	-2.73	-2.71	-2.71	-2.68	-2.68	-2.69
0.02	0.56	0.66	0.61	0.68	0.64	0.69
0.04	3.96	4.06	3.99	4.09	4.03	4.11
0.06	7.40	7.49	7.43	7.53	7.45	7.52
0.08	10.88	10.95	10.89	10.93	10.94	10.99
0.10	14.42	14.42	14.47	14.47	14.46	14.46

Solution

First, we calculate the arithmetic mean for the loading ($\bar{y}_L$) and unloading ($\bar{y}_U$) outputs at each

pressure point across the 3 cycles. Then, we find the hysteresis difference ($\Delta H = \bar{y}_U - \bar{y}_L$)

and the overall average output ($\bar{y} = \frac{\bar{y}_L + \bar{y}_U}{2}$).

x (MPa)	Avg Loading $\bar{y}_L$	Avg Unloading $\bar{y}_U$	Hysteresis Diff. ΔH	Overall Avg $\bar{y}$
0.00	-2.706	-2.693	0.013	-2.70
0.02	0.603	0.677	0.074	0.64
0.04	3.993	4.087	0.093 (Max)	4.04
0.06	7.426	7.513	0.087	7.47
0.08	10.903	10.957	0.054	10.93
0.10	14.450	14.450	0.000	14.45

From the table, the **maximum hysteresis difference** is $\Delta H_{max} \approx 0.093$ mV.

Step 2: Least Squares Fitting & Non-Linearity Error

Using the Method of Least Squares on the overall average values ($\bar{y}$), we fit the best straight line $y = a + bx$.

- By solving the normal equations (omitting intermediate steps), the fitted line is: $y = 171.5x - 2.77$
- Theoretical Full-Scale (F.S.) Output:** $y_{F.S} = y(0.10) - y(0.00) = 14.38 - (-2.77) = \mathbf{17.15}$ mV

Solution

Next, we compare the overall average ($\bar{y}$) with the theoretical value from the fitted line (y_{fitted})

to find the maximum deviation (ΔL).

- At $x = 0.00$ MPa: $y_{fitted} = -2.77$ mV, deviation = $|-2.70 - (-2.77)| = 0.07$ mV
- At $x = 0.10$ MPa: $y_{fitted} = 14.38$ mV, deviation = $|14.45 - 14.38| = 0.07$ mV
- The **maximum non-linear deviation** is $\Delta L_{max} = 0.07$ mV.

Non-linearity Error (γ_L):

$$\gamma_L = \pm \frac{\Delta L_{max}}{y_{F.S}} \times 100\% = \pm \frac{0.07}{17.15} \times 100\% = \pm \mathbf{0.41\%}$$

Step 3: Hysteresis Error Calculation

Using the maximum hysteresis difference from Step 1:

$$\gamma_H = \pm \frac{\Delta H_{max}}{y_{F.S}} \times 100\% = \pm \frac{0.093}{17.15} \times 100\% = \pm \mathbf{0.54\%}$$

Step 4: Repeatability Error Calculation

We calculate the standard deviation (S) for each calibration point using Bessel's formula. By

calculating the variance for both loading and unloading data at each point, the **maximum**

standard deviation occurs at $x = 0.02$ MPa (Loading), where $\sigma_{max} \approx 0.0404$ mV.

Assuming a confidence coefficient $k = 3$ (which corresponds to a 99.73% confidence level):

$$\gamma_R = \pm \frac{3 \cdot \sigma_{max}}{y_{F.S}} \times 100\% = \pm \frac{3 \times 0.0404}{17.15} \times 100\% = \pm \mathbf{0.71\%}$$

Sensor Dynamic Characteristics

1. Fundamental Concepts

- **Dynamic Characteristics:** Refers to the response characteristics of a sensor's output when the input signal changes over time.
- **Ideal Goal:** The output should reproduce the changes in the input signal without distortion or delay.
- **Physical Constraints:** Due to system inertia (mass) and resistance (energy dissipation), the output often lags behind the input.

2. First-Order Systems: Time Constant (τ)

- **Typical Application:** Temperature sensors (e.g., thermocouples).
- **Step Response:** $y(t) = y(\infty)[1 - e^{-t/\tau}]$.
- **Key Indicator:** The time constant τ determines response speed; a smaller τ means a faster sensor.
- **Settling Time:** After 1τ , the output reaches 63.2% of the steady-state value.

3. Second-Order Systems: Damping and Frequency

- **Typical Application:** Force sensors and piezoelectric accelerometers.
- **Key Parameters:**
 - **Natural Frequency (ω_n):** Determines the upper limit of the response speed.
 - **Damping Ratio (ζ):** Controls the oscillation and overshoot.
- **Damping States:**
 - **Underdamped ($\zeta < 1$):** Fast response but includes oscillations (ringing).
 - **Overdamped ($\zeta > 1$):** Sluggish response with no oscillation.
 - **Engineering Optimum:** Usually designed at $\zeta \approx 0.707$ to balance speed and stability.

Exercise

When the temperature of the measured medium changes abruptly from 25°C to 300°C , the temperature sensor (a typical first-order system) shows a response defined by a time constant $\tau = 120\text{ s}$.

Task: Determine the **dynamic error** of the measurement after 350 s have elapsed.

Solution

When the temperature of the measured medium changes abruptly from 25°C to 300°C , the temperature sensor (a typical first-order system) shows a response defined by a time constant $\tau = 120\text{ s}$.

Task: Determine the **dynamic error** of the measurement after 350 s have elapsed.

$$y(t) = 1 - e^{-\frac{t}{\tau}}$$
$$y(t) = 1 - e^{-\frac{t}{\tau}} = 1 - e^{-\frac{350}{120}} = 0.945$$
$$\Delta = (300 - 25) \times (1 - 0.945) = 14.88^{\circ}\text{C}$$

Exercise

A force sensor is modeled as a second-order system with a natural frequency $f_n = 800 \text{ Hz}$ and an initial damping ratio $\zeta = 0.14$.

1. Calculate the **amplitude ratio** $A(\omega)$ and the **phase angle** $\phi(\omega)$ when the sensor is used to measure a sinusoidal force with a frequency $f = 400 \text{ Hz}$.
2. If the damping ratio is adjusted to $\zeta = 0.7$, determine the new values for $A(\omega)$ and $\phi(\omega)$. Compare the results to evaluate the effect of damping.

Solution

Step 1: Identify key parameters and the frequency ratio

- Natural frequency: $f_n = 800$ Hz
- Input signal frequency: $f = 400$ Hz
- Frequency ratio: $\eta = \frac{f}{f_n} = \frac{400}{800} = 0.5$

Step 2: Case 1 — Underdamped ($\zeta = 0.14$)

Using the frequency response formulas for a second-order system:

- Amplitude Ratio $A(\omega)$:**

$$A(\omega) = \frac{1}{\sqrt{(1 - \eta^2)^2 + (2\zeta\eta)^2}}$$

$$A(\omega) = \frac{1}{\sqrt{0.75^2 + 0.14^2}} = \frac{1}{\sqrt{0.5625 + 0.0196}} \approx \mathbf{1.311}$$

- Phase Angle $\phi(\omega)$:**

$$\phi(\omega) = -\arctan\left(\frac{2\zeta\eta}{1 - \eta^2}\right)$$

$$\phi(\omega) = -\arctan\left(\frac{2 \times 0.14 \times 0.5}{1 - 0.5^2}\right) = -\arctan\left(\frac{0.14}{0.75}\right) \approx \mathbf{-10.57^\circ}$$

Step 3: Case 2 — Optimized Damping ($\zeta = 0.7$)

$$A(\omega) = \frac{1}{\sqrt{0.75^2 + 0.7^2}} = \frac{1}{\sqrt{0.5625 + 0.49}} \approx \mathbf{0.975}$$

$$\phi(\omega) = -\arctan\left(\frac{2 \times 0.7 \times 0.5}{0.75}\right) = -\arctan\left(\frac{0.7}{0.75}\right) \approx \mathbf{-43.03^\circ}$$